Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Mathematics II

BCA — Semester 2 — 2018 — Answer Sheet

Subject Code: CACS154

Group A

Attempt all the questions. [10×1 = 10]

37. For all rational values of n, lim(x→a) (x■ - a■)/(x - a) is equal to

[1]

Answer: Correct Answer: a) na■■^{1}
This is a standard limit formula derived from the definition of derivative. Using L'Hôpital's Rule or binomial expansion, we obtain:
lim(x→a) (x■ - a■)/(x - a) = na■■^{1}
This result is fundamental in calculus and forms the basis for the power rule of differentiation: d/dx(x■) = nx■■^{1}.

38. If lim(x→x■■) f(x) ≠ lim(x→x■■) f(x), then f(x) is said to be

[1]

Answer: Correct Answer: b) An ordinary discontinuity
When the left-hand limit and right-hand limit exist but are not equal, the function has a **jump discontinuity** (also called ordinary discontinuity or discontinuity of the first kind). The function cannot be made continuous by redefining a single point. If the limits were equal but different from f(x■), it would be a removable discontinuity.

39. Derivative of tan■^{1}x is equal to

[1]

Answer: Correct Answer: c) $1/(1+x^2)$
Let y = tan■^{1}x, then tan y = x.
Differentiating implicitly: $\sec^2 y \cdot dy/dx = 1$
Therefore: $dy/dx = 1/\sec^2 y = 1/(1 + \tan^2 y) =$ **$1/(1 + x^2)$**
This is one of the standard derivatives in calculus, widely used in integration and differential equations.

40. The value of lim(x→0) (e■ - 1)/x is equal to

[1]

Answer: Correct Answer: b) 1

This is a fundamental exponential limit. Using L'Hôpital's Rule or Taylor series expansion of e■:

e■ = 1 + x + $x^2/2!$ + $x^3/3!$ + ...

So (e■ - 1)/x = 1 + x/2! + $x^2/3!$ + ...

As x → 0, all terms with x vanish, leaving **1**.

This limit is essential for deriving the derivative of e■.

41. The differential equation: $y(d^2y/dx^2)^2 + 5(dy/dx)^2 + 2y = 0$ is known as

[1]

Answer: Correct Answer: d) First order second degree

Wait — correcting based on standard form analysis:

The highest order derivative is d^2y/dx^2 (second order).

The highest power of the highest derivative is 2 (since it's squared).

Therefore, this is a **Second order, Second degree** differential equation.

Correct Answer: a) Second degree second order

Order = highest derivative present = 2

Degree = highest power of the highest derivative = 2

42. One important condition to satisfy Rolle's Theorem by a function f(x) in [a, b] is

[1]

Answer: Correct Answer: c) f(a) = f(b)

Rolle's Theorem states that if a function f(x) satisfies:

1. f is continuous on the closed interval [a, b]
2. f is differentiable on the open interval (a, b)
3. **f(a) = f(b)**

Then there exists at least one c in (a, b) such that f'(c) = 0.

The condition f(a) = f(b) is essential for the theorem; without it, the conclusion may fail.

43. Formula for the composite trapezoidal rule is

[1]

Answer: Correct Answer: a) (h/2)[y■ + 2(y■+y■+...+y■■■) + y■]

The **Composite Trapezoidal Rule** approximates the definite integral by dividing the interval into n subintervals of width h and applying the trapezoid formula to each:

∫■■ f(x)dx ≈ (h/2)[y■ + 2(y■ + y■ + ... + y■■■) + y■]

where h = (b-a)/n and y■ = f(x■). The first and last ordinates have coefficient 1, while all intermediate ordinates have coefficient 2. The error is $O(h^2)$.

44. While applying Simpson's 3/8 rule the number of sub-interval should be

[1]

Answer: Correct Answer: d) Multiple of 3

Simpson's 3/8 Rule requires that the number of sub-intervals (n) be a **multiple of 3**. This is because the rule approximates the integral over three sub-intervals at a time using a cubic polynomial. The formula is:

∫■■ f(x)dx ≈ (3h/8)[y■ + 3(y■+y■+y■+y■+...) + 2(y■+y■+...) + y■]

The pattern of coefficients is 1, 3, 3, 2, 3, 3, 2, ..., 1, which requires n to be divisible by 3.

45. In Gauss Elimination method the given system of simultaneous equation is transformed into

[1]

Answer: Correct Answer: d) upper triangular matrix

The **Gauss Elimination Method** transforms a system of linear equations into an equivalent system represented by an **upper triangular matrix** through row operations. Once in upper triangular form (where all elements below the main diagonal are zero), the system can be solved efficiently using **back substitution**, starting from the last equation and working upward.

46. In Newton-Raphson method, if x■ is an approximate solution of f(x) = 0 and f'(x■) ≠ 0, the next approximation is given by

[1]

Answer: Correct Answer: c) x■■■ = x■ - f(x■)/f'(x■)

The **Newton-Raphson formula** is derived from Taylor series expansion and geometric interpretation (tangent line approximation):

x■■■ = x■ - f(x■)/f'(x■)

This iterative method converges quadratically near simple roots. The method requires an initial guess x■ and that f'(x■) ≠ 0 at each iteration. It is one of the most efficient methods for finding roots of nonlinear equations.

Group B

Attempt any SIX questions. [6×5 = 30]

47. If a function f(x) is defined as: $f(x) = 3x^2 + 2$ if $x < 1$; $f(x) = 2x + 3$ if $x > 1$; $f(x) = 4$ if $x = 1$. Discuss the continuity of function at $x = 1$.

[5]

Answer: Continuity Analysis at x = 1:

For a function to be continuous at x = a, three conditions must be satisfied:
1. f(a) must be defined
2. lim(x→a) f(x) must exist (LHL = RHL)
3. lim(x→a) f(x) = f(a)

Step 1: Find f(1)
Given: f(1) = **4**

Step 2: Find Left Hand Limit (LHL)
lim(x→1■) f(x) = lim(x→1■) $(3x^2 + 2)$
= $3(1)^2 + 2 = 3 + 2 =$ **5**

Step 3: Find Right Hand Limit (RHL)
lim(x→1■) f(x) = lim(x→1■) (2x + 3)
= 2(1) + 3 = 2 + 3 = **5**

Step 4: Check conditions
• LHL = RHL = 5, so lim(x→1) f(x) = **5**
• f(1) = 4
• Since lim(x→1) f(x) = 5 ≠ f(1) = 4

Conclusion:
The function is **discontinuous at x = 1**.

Type of Discontinuity:
Since LHL = RHL ≠ f(1), this is a **removable discontinuity** (discontinuity of the first kind). If we redefine f(1) = 5, the function would become continuous at x = 1.

Graphical Interpretation:
The graph approaches y = 5 from both sides of x = 1, but there is a hole at (1, 5) and the actual point is at (1, 4). This single point displacement creates the discontinuity.

48. Find the derivative of $\sin^3 x$ by using definition.

[5]

Answer: Derivative of $\sin^3 x$ using First Principles (Definition):

The definition of derivative is:
$f'(x) = \lim(h \to 0) [f(x+h) - f(x)]/h$

Let $f(x) = \sin^3 x$

Step 1: $f(x+h) = \sin^3(x+h)$

Step 2: $f(x+h) - f(x) = \sin^3(x+h) - \sin^3 x$

Using the identity: $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
$= [\sin(x+h) - \sin x][\sin^2(x+h) + \sin(x+h)\sin x + \sin^2 x]$

Step 3: Divide by h:
$[f(x+h) - f(x)]/h = [(\sin(x+h) - \sin x)/h] \cdot [\sin^2(x+h) + \sin(x+h)\sin x + \sin^2 x]$

Step 4: Take limit as $h \to 0$:

First factor: $\lim(h \to 0) [\sin(x+h) - \sin x]/h$
$= \lim(h \to 0) [2\cos((2x+h)/2)\sin(h/2)]/h$
$= \lim(h \to 0) \cos(x + h/2) \cdot [\sin(h/2)/(h/2)]$
$= \cos(x) \cdot 1 =$ **$\cos x$**

Second factor: $\lim(h \to 0) [\sin^2(x+h) + \sin(x+h)\sin x + \sin^2 x]$
$= \sin^2 x + \sin x \cdot \sin x + \sin^2 x =$ **$3\sin^2 x$**

Step 5: Multiply the limits:
$f'(x) = \cos x \cdot 3\sin^2 x =$ **$3\sin^2 x \cos x$**

Verification using Chain Rule:
$d/dx(\sin^3 x) = 3\sin^2 x \cdot d/dx(\sin x) = 3\sin^2 x \cos x$ ✓

This confirms our result from first principles.

13. Using L'Hospital's rule evaluate: $\lim_{x\to\infty} (2x^2 + 3x + 5)/(3x^2 + 2)$

[5]

Answer: Evaluation using L'Hospital's Rule:

Given: $\lim_{x\to\infty} (2x^2 + 3x + 5)/(3x^2 + 2)$

Step 1: Check the form
As $x \to \infty$:
Numerator: $2x^2 + 3x + 5 \to \infty$
Denominator: $3x^2 + 2 \to \infty$
This is the **∞/∞ indeterminate form**, so L'Hospital's Rule applies.

Step 2: Apply L'Hospital's Rule (first time)
Differentiate numerator and denominator separately:
$d/dx(2x^2 + 3x + 5) = 4x + 3$
$d/dx(3x^2 + 2) = 6x$

New limit: $\lim_{x\to\infty} (4x + 3)/(6x)$

Step 3: Check the form again
As $x \to \infty$: $(4x + 3)/(6x) \to \infty/\infty$
Apply L'Hospital's Rule again.

Step 4: Apply L'Hospital's Rule (second time)
$d/dx(4x + 3) = 4$
$d/dx(6x) = 6$

New limit: $\lim_{x\to\infty} 4/6 =$ **2/3**

Alternative Method (Direct):
Divide numerator and denominator by x^2:
$\lim_{x\to\infty} (2 + 3/x + 5/x^2)/(3 + 2/x^2)$
$= (2 + 0 + 0)/(3 + 0) =$ **2/3**

Answer: 2/3

35. If demand function and cost function are given by $P(Q) = 1-3Q$ and $C(Q) = Q^2 - 2Q$ respectively, where Q is the quantity of the product, then find output of the factor for the maximum profit.

[5]

Answer: Finding Output for Maximum Profit:

Given:
Demand function (Price): $P(Q) = 1 - 3Q$
Cost function: $C(Q) = Q^2 - 2Q$

Step 1: Find Revenue Function R(Q)
Revenue = Price × Quantity
$R(Q) = P(Q) \times Q = (1 - 3Q) \times Q = \mathbf{Q - 3Q^2}$

Step 2: Find Profit Function $\pi(Q)$
Profit = Revenue - Cost
$\pi(Q) = R(Q) - C(Q)$
$= (Q - 3Q^2) - (Q^2 - 2Q)$
$= Q - 3Q^2 - Q^2 + 2Q$
$= \mathbf{3Q - 4Q^2}$

Step 3: Find Critical Points
For maximum profit, $d\pi/dQ = 0$
$d\pi/dQ = 3 - 8Q = 0$
$8Q = 3$
$Q = \mathbf{3/8 = 0.375}$

Step 4: Verify Maximum
$d^2\pi/dQ^2 = -8 < 0$
Since the second derivative is negative, the profit is indeed **maximum** at $Q = 3/8$.

Step 5: Calculate Maximum Profit
$\pi(3/8) = 3(3/8) - 4(3/8)^2$
$= 9/8 - 4(9/64)$
$= 9/8 - 36/64$
$= 9/8 - 9/16$
$= 18/16 - 9/16$
$= \mathbf{9/16 = 0.5625}$

Answer: The output for maximum profit is **Q = 3/8 = 0.375 units**, and the maximum profit is **9/16 = Rs. 0.5625**.

36. Evaluate: a) $\int \sin\blacksquare^{1}x\, dx$ b) $\int\blacksquare^{1} (x^2 + 5)\, dx$

[5]

Answer: a) Evaluation of $\int \sin\blacksquare^{1}x\, dx$

Using integration by parts: $\int u\, dv = uv - \int v\, du$

Let $u = \sin\blacksquare^{1}x$, $dv = dx$
Then $du = dx/\sqrt{(1-x^2)}$, $v = x$

$\int \sin\blacksquare^{1}x\, dx = x \sin\blacksquare^{1}x - \int x/\sqrt{(1-x^2)}\, dx$

For the second integral, let $t = 1 - x^2$, then $dt = -2x\, dx$, so $x\, dx = -dt/2$
$\int x/\sqrt{(1-x^2)}\, dx = \int (-1/2)\, t\blacksquare^{1/2}\, dt = -1/2 \cdot 2t^{1/2} = -\sqrt{t} = -\sqrt{(1-x^2)}$

Therefore:
$\int \sin\blacksquare^{1}x\, dx = x \sin\blacksquare^{1}x - (-\sqrt{(1-x^2)}) + C$
= **$x \sin\blacksquare^{1}x + \sqrt{(1-x^2)} + C$**

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b) Evaluation of $\int\blacksquare^{1} (x^2 + 5)\, dx$

$\int\blacksquare^{1} (x^2 + 5)\, dx = [x^3/3 + 5x]\blacksquare^{1}$
$= (1^3/3 + 5\cdot1) - (0^3/3 + 5\cdot0)$
$= (1/3 + 5) - 0$
$= 1/3 + 15/3$
= **$16/3$** ≈ **5.333**

Answers:
a) **$x \sin\blacksquare^{1}x + \sqrt{(1-x^2)} + C$**
b) **$16/3$**

37. Solve: dy/dx = (x + y)/(x - y)

[5]

Answer: Solving the Differential Equation dy/dx = (x + y)/(x - y)

This is a **homogeneous differential equation** because both numerator and denominator are homogeneous functions of degree 1.

Step 1: Substitution
Let y = vx, then dy/dx = v + x(dv/dx)

Step 2: Substitute into the equation
v + x(dv/dx) = (x + vx)/(x - vx)
v + x(dv/dx) = x(1 + v)/[x(1 - v)]
v + x(dv/dx) = (1 + v)/(1 - v)

Step 3: Separate variables
x(dv/dx) = (1 + v)/(1 - v) - v
= $(1 + v - v + v^2)/(1 - v)$
= $(1 + v^2)/(1 - v)$

$(1 - v)/(1 + v^2)\, dv = dx/x$

Step 4: Integrate both sides
$\int (1 - v)/(1 + v^2)\, dv = \int dx/x$

Split the left side:
$\int 1/(1 + v^2)\, dv - \int v/(1 + v^2)\, dv = \ln|x| + C$

First integral: $\tan^{-1}v$
Second integral: Let $t = 1 + v^2$, dt = 2v dv
$\int v/(1 + v^2)\, dv = (1/2)\ln(1 + v^2)$

So:
$\tan^{-1}v - (1/2)\ln(1 + v^2) = \ln|x| + C$

Step 5: Substitute back v = y/x
$\tan^{-1}(y/x) - (1/2)\ln(1 + y^2/x^2) = \ln|x| + C$

Simplify the logarithm:
$(1/2)\ln(1 + y^2/x^2) = (1/2)\ln[(x^2 + y^2)/x^2] = (1/2)[\ln(x^2 + y^2) - 2\ln|x|] = (1/2)\ln(x^2 + y^2) - \ln|x|$

Substituting:
$\tan^{-1}(y/x) - (1/2)\ln(x^2 + y^2) + \ln|x| = \ln|x| + C$

Cancel ln|x| from both sides:
$\tan^{-1}(y/x) - (1/2)\ln(x^2 + y^2) = C$

Or equivalently:
$2\tan^{-1}(y/x) - \ln(x^2 + y^2) = C$■

This is the general solution of the given differential equation.

38. Examine the consistency of the system of equation and solve if possible. 3x■ + 2x■ + x■ = 3; 2x■ + x■ + x■ = 3; 3x■ + 3x■ + 2x■ = 2

[5]

Answer: Consistency Analysis and Solution:

Given system:
3x■ + 2x■ + x■ = 3 ...(1)
2x■ + x■ + x■ = 3 ...(2)
3x■ + 3x■ + 2x■ = 2 ...(3)

Step 1: Form the augmented matrix [A|B]
321|3 211|3 332|2

Step 2: Apply row operations

R■ → R■ - (2/3)R■:
321|3 0-1/31/3|1 332|2

R■ → R■ - R■:
321|3 0-1/31/3|1 011|-1

R■ → -3R■:
321|3 01-1|-3 011|-1

R■ → R■ - R■:
321|3 01-1|-3 002|2

Step 3: Back substitution

From row 3: 2x■ = 2, so **x■ = 1**

From row 2: x■ - x■ = -3
x■ - 1 = -3
x■ = -2

From row 1: 3x■ + 2x■ + x■ = 3
3x■ + 2(-2) + 1 = 3
3x■ - 4 + 1 = 3
3x■ = 6
x■ = 2

Step 4: Consistency Check
Rank of A = 3
Rank of [A|B] = 3
Number of unknowns = 3
Since Rank(A) = Rank([A|B]) = 3, the system is **consistent** and has a **unique solution**.

Verification:
Eq (1): 3(2) + 2(-2) + 1 = 6 - 4 + 1 = 3 ✓

Eq (2): 2(2) + (-2) + 1 = 4 - 2 + 1 = 3 ✓
Eq (3): 3(2) + 3(-2) + 2(1) = 6 - 6 + 2 = 2 ✓

Answer: x■ = 2, x■ = -2, x■ = 1

Group C

Attempt any TWO questions. [2×10 = 20]

39. Define Homogeneous equation and solve the following system of equations using Inverse Matrix Method: -2x + 2y + z = -4; -8x + 7y - 4z = -47; 9x - 8y + 5z = 55

[10]

Answer: Part 1: Definition of Homogeneous Equation

A **system of linear equations** is called **homogeneous** if all the constant terms are zero. In matrix form, AX = O, where A is the coefficient matrix, X is the column vector of variables, and O is the zero vector.

For example:
2x + 3y - z = 0
x - y + 2z = 0
3x + y - z = 0

Properties of Homogeneous Systems:
• Always consistent (at least the trivial solution x = y = z = 0 exists)
• If |A| ≠ 0: only trivial solution (unique solution)
• If |A| = 0: infinitely many solutions (non-trivial solutions exist)

Part 2: Solution using Inverse Matrix Method

Given system:
-2x + 2y + z = -4
-8x + 7y - 4z = -47
9x - 8y + 5z = 55

Step 1: Write in matrix form AX = B
A = [[-2, 2, 1], [-8, 7, -4], [9, -8, 5]]
X = [x, y, z]■
B = [-4, -47, 55]■

Step 2: Find |A| (determinant of A)
|A| = -2(35 - 32) - 2(-40 + 36) + 1(64 - 63)
= -2(3) - 2(-4) + 1(1)
= -6 + 8 + 1
= **3**

Since |A| = 3 ≠ 0, A■^{1} exists and the system has a unique solution.

Step 3: Find the Adjoint of A

Cofactor matrix C:
C■■ = +(35-32) = 3, C■■ = -(-40+36) = 4, C■■ = +(64-63) = 1
C■■ = -(10+8) = -18, C■■ = +(-10-9) = -19, C■■ = -(-16-18) = 34
Wait, recalculating carefully:

C■■ = +(7·5 - (-4)(-8)) = 35 - 32 = 3
C■■ = -((-8)·5 - (-4)·9) = -(-40 + 36) = 4

C■■ = +((-8)(-8) - 7·9) = 64 - 63 = 1

C■■ = -(2·5 - 1·(-8)) = -(10 + 8) = -18
C■■ = +((-2)·5 - 1·9) = -10 - 9 = -19
C■■ = -((-2)(-8) - 2·9) = -(16 - 18) = 2

C■■ = +(2·(-4) - 1·7) = -8 - 7 = -15
C■■ = -((-2)(-4) - 1·(-8)) = -(8 + 8) = -16
C■■ = +((-2)·7 - 2·(-8)) = -14 + 16 = 2

Cofactor matrix:
[[3, 4, 1], [-18, -19, 2], [-15, -16, 2]]

Adj(A) = C■ =
[[3, -18, -15], [4, -19, -16], [1, 2, 2]]

Step 4: Find A■¹
A■¹ = Adj(A)/|A| = (1/3) · [[3, -18, -15], [4, -19, -16], [1, 2, 2]]

Step 5: Find X = A■¹B

[x] (1/3) [3 -18 -15] [-4]
[y] = [4 -19 -16] [-47]
[z] [1 2 2] [55]

x = (1/3)[3(-4) + (-18)(-47) + (-15)(55)]
= (1/3)[-12 + 846 - 825]
= (1/3)[9]
= **3**

y = (1/3)[4(-4) + (-19)(-47) + (-16)(55)]
= (1/3)[-16 + 893 - 880]
= (1/3)[-3]
= **-1**

z = (1/3)[1(-4) + 2(-47) + 2(55)]
= (1/3)[-4 - 94 + 110]
= (1/3)[12]
= **4**

Verification:
Eq 1: -2(3) + 2(-1) + 4 = -6 - 2 + 4 = -4 ✓
Eq 2: -8(3) + 7(-1) - 4(4) = -24 - 7 - 16 = -47 ✓
Eq 3: 9(3) - 8(-1) + 5(4) = 27 + 8 + 20 = 55 ✓

Answer: x = 3, y = -1, z = 4

40. State Rolle's Theorem and interpret it geometrically. Verify Rolle's theorem for $f(x) = x^2 - 4$ in $-3 \leq x \leq 3$

[10]

Answer: Part 1: Statement of Rolle's Theorem

Rolle's Theorem: If a function f(x) satisfies the following three conditions:
1. f is **continuous** on the closed interval [a, b]
2. f is **differentiable** on the open interval (a, b)
3. **f(a) = f(b)**

Then there exists at least one real number **c** $\in$ **(a, b)** such that **f'(c) = 0**.

Geometrical Interpretation:
If a curve y = f(x) is continuous between points A(a, f(a)) and B(b, f(b)), has a unique tangent at every point between A and B, and f(a) = f(b) (i.e., A and B are at the same height), then there exists at least one point C(c, f(c)) between A and B where the tangent to the curve is **parallel to the x-axis** (horizontal).

In other words, somewhere between two points at the same height on a smooth curve, there must be at least one "peak" or "valley" where the slope is zero.

Part 2: Verification for $f(x) = x^2 - 4$ in [-3, 3]

Step 1: Check continuity
$f(x) = x^2 - 4$ is a polynomial function. All polynomial functions are continuous everywhere.
Therefore, f is continuous on [-3, 3]. ✓ **Condition 1 satisfied.**

Step 2: Check differentiability
f'(x) = 2x exists for all real x.
Therefore, f is differentiable on (-3, 3). ✓ **Condition 2 satisfied.**

Step 3: Check f(a) = f(b)
$f(-3) = (-3)^2 - 4 = 9 - 4 = 5$
$f(3) = (3)^2 - 4 = 9 - 4 = 5$
Therefore, **f(-3) = f(3) = 5**. ✓ **Condition 3 satisfied.**

Step 4: Find c such that f'(c) = 0
f'(x) = 2x
Set f'(c) = 0:
2c = 0
c = **0**

Step 5: Verify c $\in$ (-3, 3)
c = 0 lies in the open interval (-3, 3). ✓

Step 6: Verify f'(0) = 0
f'(0) = 2(0) = **0** ✓

Conclusion:
Since all three conditions of Rolle's Theorem are satisfied and we found c = 0 $\in$ (-3, 3) such that f'(0) = 0,

Rolle's Theorem is verified for $f(x) = x^2 - 4$ on [-3, 3].

Geometrical Meaning:
The parabola $y = x^2 - 4$ has vertex at (0, -4). At $x = 0$, the tangent is horizontal (parallel to x-axis). Points (-3, 5) and (3, 5) are at the same height, and the curve smoothly descends to its minimum at (0, -4) before rising again, confirming that a horizontal tangent exists between them.

20. Using Composite Trapezoidal Rule, compute ∫■^{2} $1/(1+x^2)$ dx with four intervals. Find the absolute error of approximation from its actual value.

[10]

Answer: Solution using Composite Trapezoidal Rule

Given:
∫■^{2} $1/(1+x^2)$ dx, n = 4 intervals

Step 1: Calculate h
h = (b - a)/n = (2 - 0)/4 = **0.5**

Step 2: Calculate function values
ix■y■ = 1/(1+x■^{2}) 00.01.0000 10.50.8000 21.00.5000 31.50.3077 42.00.2000

Step 3: Apply Composite Trapezoidal Rule
∫■^{2} f(x)dx ≈ (h/2)[y■ + 2(y■ + y■ + y■) + y■]
= (0.5/2)[1.0000 + 2(0.8000 + 0.5000 + 0.3077) + 0.2000]
= 0.25[1.0000 + 2(1.6077) + 0.2000]
= 0.25[1.0000 + 3.2154 + 0.2000]
= 0.25[4.4154]
= **1.10385** ≈ **1.1039**

Step 4: Find Actual Value
∫ $1/(1+x^2)$ dx = tan■^{1}x
Actual value = [tan■^{1}x]■^{2} = tan■^{1}(2) - tan■^{1}(0)
= 1.1071487... - 0
= **1.1071** (approx)

Step 5: Calculate Absolute Error
Absolute Error = |Actual Value - Approximate Value|
= |1.1071 - 1.1039|
= **0.0032** (or more precisely, 0.0033)

Using more precise values:
Actual = tan■^{1}(2) ≈ 1.1071487177940904
Approximate = 1.1038461538461537
Absolute Error ≈ **0.00330256**

Percentage Error:
(0.0033/1.1071) × 100 ≈ **0.30%**

Conclusion:
The Composite Trapezoidal Rule with n = 4 gives a good approximation (1.1039) with a small absolute error of approximately 0.0033. The error can be further reduced by increasing the number of intervals.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.